The

ASTER RETAIL HOLDINGS

DATA ANALYTICS REPORT

A comprehensive documentation of the data cleaning, transformation and preparation process carried out using Python (pandas) and Excel/Power BI for Aster Retail Holdings.

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1. Overview

The Aster Retail Holdings (London) project showcases a complete data analytics workflow from raw data cleaning to final dashboard reporting using Python, Excel, Power BI, Git and GitHub. The goal was to analyze historical sales data, identify key revenue drivers and present insights, for the year 2022, through clear visual storytelling.

The process began with cleaning and transforming the raw CSV file using Python (pandas) like fixing date formats, handling missing values, correcting product categories, renaming columns and creating new fields such as Total Sales, which was initially missing in the raw dataset. The cleaned dataset was saved in both CSV and Excel formats for analysis and version-controlled through GitHub, ensuring proper documentation and reproducibility.

In Excel, initial Exploratory Data Analysis (EDA) was conducted using PivotTables to identify early trends and define key metrics such as Total Sales, Sales by Category, Performance Trends and Total Sales by Country for each category.

The cleaned data was then imported into Power BI where DAX measures and dashboards were built. Visuals included category-wise and regional sales charts, AOV and KPI cards and interactive slicers for dynamic analysis. Each page emphasized clarity and storytelling — highlighting performance patterns, regional trends, and customer insights.

The project deliverables included a cleaned dataset, functional Power BI dashboard and a summary report with key insights and screenshots, all managed through GitHub.

2. Objective of the Project

The objective of this project was to perform a complete end-to-end sales data analysis for Aster Retail Holdings, covering every stage from data cleaning to dashboard presentation. The project aimed to ensure data accuracy, analytical clarity and insightful storytelling through visualization techniques.

Key Objectives and Features:

- Find a real-world dataset from the Web.
- Document and execute all data cleaning and pre-processing steps, including handling missing data, formatting dates etc.
- Conduct Exploratory Data Analysis (EDA) in Excel using PivotTables to uncover trends and establish initial Key Performance Indicators (KPIs).
- Develop interactive Power BI dashboards with synchronized slicers, drill-down functionality and visual storytelling to enhance analytical understanding.
- Clearly define and measure core KPIs such as Total Sales, Sales by Category, Average Order Value (AOV) and Distinct Customer Count.
- Polish and finalize the dashboard design with a proper, business-oriented layout suitable for presentation.
- Create a comprehensive summary report including key findings, visual documentation and strategic recommendations.

Final Deliverables:

- Completed and fully interactive Power BI Dashboard.
- Summary Report highlighting key insights and recommendations.
- Attaching supporting screenshots, documentation.

3. Tools and Libraries Used

Tool / Library	Purpose
pandas (Python)	Data Cleaning and Transformation
openpyxl (Python)	Excel File Exportation
Microsoft Excel	Pivot and Exploratory Data Analysis
Power BI	Data Visualization and Dashboard Creation
GitHub	Version-Control and Documentation

4. Steps Undertaken

1. Cloned the original repository and created a structured directory for raw and cleaned data, python scripts, dashboard and report.

2. Conducted Exploratory Data Analysis (EDA) for Sales by Category, Sales by Month and Sales by Country for each Category in Excel.

3. Basic visualisation charts were created along with PivotTables for each of the EDA methods.

4. Imported the resulting raw CSV file into Python for pre- processing.

5. Conducted initial inspection: verified shape, data types and sample records.

6. Converted ‘order_date’ to datetime format and standardized text columns.

7. Checked for missing values and handled null entries appropriately.

8. Standardized column naming conventions (lowercase, underscore separation).

9. Verified duplicates in ‘customer_id’ and ‘product_id’ columns.

10. Corrected product categories using a pre-defined mapping dictionary.

11. Added a new calculated ‘sales’ column using quantity × unit_price.

12. Exported cleaned data to both CSV and Excel formats for Power BI import.

5. Summary of Exploratory Data Analysis (Excel)

1. Imported the cleaned dataset into Excel using Get Data → From Text/CSV.

2. Verified data consistency through format checks (dates, numeric fields text alignment).

3. Created PivotTables to analyze Total Sales, Sales by Category and Monthly Trends.

4. Added slicers for dynamic filtering by Product Category and Region.

5. Designed charts and visual summaries (column, line and pie charts) to represent KPIs.

6. Compiled findings into an interactive Excel dashboard with labelled insights and screenshots for documentation.

Pivot Table 1

Total Sales (£) by Category	
Categories	**Total (£)**
Accessories	203010
Clothing	231421
Electronics	297359
Grand Total	**731790**

Pivot Table 1 Insights

1. The Initial Analysis of sales data indicates that the combined revenue from Clothing, Electronics and Accessories totalled for about £730K for the year 2022, reflecting the overall performance across all product categories.

2. The Electronics category recorded the highest sales, close to £300K, while the Accessories category reported the lowest sales at £203K.

Pivot Table 2

Total Sales (£) by Months	
Months (2022)	Total (£)
Jan	54020
Feb	61316
Mar	54013
Apr	65379
May	64927
Jun	64948
Jul	61130
Aug	96241
Sep	66313
Oct	51339
Nov	46723
Dec	45441
Grand Total	731790

Dark Green	Highest
Green	> 60000
Red	<60000

Pivot Table 2 Insights

1. August recorded the highest total sales of £96,241, highlighted in dark green.

2. Months such as April (£65,379), May (£64,927), June (£64,948), and September (£66,313) consistently stayed above £60,000, indicating a stable Q2–Q3 performance.

3. The months January, March, October, November and December had totals below £60,000, marked in red, signalling reduced demand or end-of-year slowdowns.

4. December (£45,441) recorded the lowest monthly sales.

Pivot Table 3

Total Sales (£) by Country				
Total by Country	**Categories**			
Countries	**Accessories**	**Clothing**	**Electronics**	**Grand Total**
Australia	28317	41751	44599	114667
Canada	43430	39613	55185	138228
France	33617	38676	31497	103790
Germany	30277	46001	51717	127995
UK	28309	26520	53620	108449
USA	39060	38860	60741	138661
Grand Total	**203010**	**231421**	**297359**	**731790**

Average of products		40655
Green	> 40,655	
Yellow	>30,000 & <40,655	
Red	<30,000	
Highest Sales by Country	USA	138661
Lowest Sales by Country	France	103790
Highest Selling Item	Electronics (USA)	60741
Lowest Selling Item	Clothing (UK)	26520

Pivot Table 3 Insights

1. Electronics category leads with £297,359, contributing over 40% of total sales, followed by Clothing (£231,421) and Accessories (£203,010).

2. The USA records the highest total sales (£138,661), outperforming all other markets largely driven by the electronics category.

3. France reports the lowest overall sales (£103,90).

4. The highest selling category was found to be Electronics in the USA, and the lowest was found to be Clothings in the UK.

6. Summary of Python Operations

A custom Python script was developed using Pandas and OpenPyXL to perform systematic data cleaning, transformation and exportation of the Aster Retail Holdings (London) dataset. The script ensured that the data was consistent, accurate and analysis-ready for Power BI (Visualisation) use.

The process began by loading and validating the dataset, checking its structure, shape, and sample rows to confirm successful import. Product categories were standardized through a mapping dictionary, ensuring correct alignment between product names (e.g., T-shirt, Tablet, Watch) and their respective categories (Clothing, Electronics, Accessories). Any and all inconsistencies were resolved using a reverse mapping function that automated category correction.

Data cleaning steps included:

- Date Standardization: Converted order_date into proper datetime format using to_datetime() with error coercion to handle invalid entries.
- Null & Duplicate Checks: Assessed for missing values and duplicate customer_id and product_id entries, ensuring no redundant or incomplete records impacted analysis.
- New Calculated Field: Introduced a new column sales as the product of quantity and unit_price for accurate revenue computation.
- Column Normalization: Converted all column names to lowercase and replaced spaces with underscores for uniformity and compatibility across tools.
- Reverse Mapping for Category Correction: Created a reverse mapping dictionary where each product name was automatically linked to its correct category. This ensured consistent classification across the dataset and eliminated human error in category labelling.
- After successful validation, the cleaned dataset was exported in both CSV and Excel formats for broader usability. The use of OpenPyXL enabled smooth Excel integration for pivot analysis and further visualization in Power BI.

Python Output Terminal

```
File Loaded Successfully
Shape of the file =  (500, 11)
First 5 rows
   order_id  customer_id  product_id product_name     category  unit_price  quantity  order_date  customer_age customer_gender    country
0         1          150           9       Tablet  Electronics         611         1  14-11-2022            62          Female    Germany
1         2          176          36   Smartphone  Electronics         751         4  06-10-2022            23            Male     Canada
2         3           37          22       Jacket  Accessories         821         4  07-05-2022            54            Male         UK
3         4          143          42        Shoes  Electronics         395         4  09-05-2022            50            Male    Germany
4         5          163          26       Jacket  Accessories         301         1  27-02-2022            39            Male  Australia
object
d:\Git\Infotact-Solution-Project-1\Scripts\Cleaned_data.py:34: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specif
ied. Pass `dayfirst=True` or specify a format to silence this warning.
  df['order_date'] = pd.to_datetime(df['order_date'], errors='coerce')
datetime64[ns]
Missing values per column :
order_id           0
customer_id        0
product_id         0
product_name       0
category           0
unit_price         0
quantity           0
order_date         0
customer_age       0
customer_gender    0
country            0
dtype: int64
   order_id  customer_id  product_id product_name     category  unit_price  quantity order_date  customer_age customer_gender    country  sales
0         1          150           9       Tablet  Electronics         611         1 2022-11-14            62          Female    Germany    611
1         2          176          36   Smartphone  Electronics         751         4 2022-10-06            23            Male     Canada   3004
2         3           37          22       Jacket     Clothing         821         4 2022-05-07            54            Male         UK   3284
3         4          143          42        Shoes     Clothing         395         4 2022-05-09            50            Male    Germany   1580
4         5          163          26       Jacket     Clothing         301         1 2022-02-27            39            Male  Australia    301
Total Duplicate Customer IDs :  317
Total Duplicate Product IDs :  450
Data types :  order_id                int64
customer_id                int64
product_id                 int64
product_name              object
category                  object
unit_price                 int64
quantity                   int64
order_date        datetime64[ns]
customer_age               int64
customer_gender           object
country                   object
sales                      int64
dtype: object
Data cleaned and saved successfully to: data/cleaned/Aster Retail Holdings Cleaned.csv
```

7. Power BI Integration and Key Insights

1. Imported the cleaned Excel dataset into Power BI and established data model relationships between key fields (e.g., Product, Category, Region and Customer).

2. Developed essential DAX measures such as Total Sales, Unique Products, Distinct Customer IDs, Average Order Value (AOV) and Sales Contribution by Category to enable deeper analytical insights.

3. Constructed multiple interactive visualizations including category-wise sales pie charts, regional performance maps, monthly trend lines and customer segmentation bubble charts for comprehensive performance tracking.

4. Integrated synchronized slicers and drill-through filters (for year, region and category) to enhance report interactivity and allow dynamic, user-driven exploration of KPIs.

5. Applied consistent colour themes, data labelling and layout design for readability and storytelling.

6. Enabled data storytelling through page navigation, with dedicated views for overall sales performance, customer behaviour analysis and geographic insights, ensuring actionable outcomes for business decision-making.

7. Validated dashboard accuracy by cross-checking DAX outputs with original Excel calculations to ensure reliability and consistency across all metrics.

8. Published and shared the Power BI report to GitHub, documenting version history, visuals and insights for transparent collaboration and review.

1. Executive Summary (Dashboard Overview)

The Executive Summary page serves as the central hub of the Power BI report, offering a holistic view of Aster Retail Holdings' sales performance. Key metrics such as Total Sales (£732K), Total Quantity (1,287), Total Products (10), Regions (6) and Average Order Value (1.46K) provide quick, quantifiable insights.

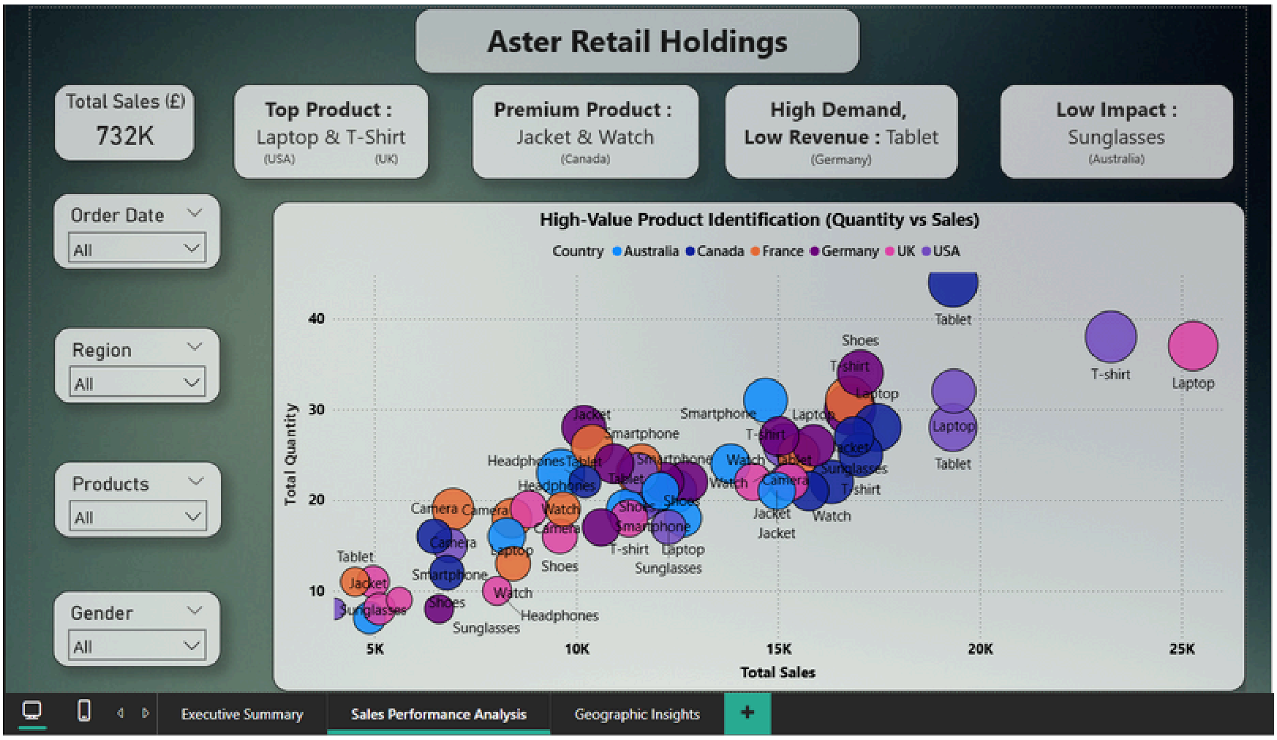
- Performance Trend: The monthly sales trend indicates strong performance during August (£96K), representing the peak sales month possibly due to increased purchase of products like Sunglasses and electronics such as Air Conditioners. It could also mean a mid-year discount campaign. Sales then gradually declined towards year-end, implying seasonal patterns or reduced consumer demand.

- Category Analysis: Electronics dominated with 40.63% of total revenue, followed by Clothing (31.62%) and Accessories (27.74%). This breakdown highlights the business's reliance on electronic products for revenue generation.

- Regional Sales: The USA (£139K) and Canada (£138K approx.) emerged as top-performing regions, while France (£104K approx.) recorded the lowest sales, suggesting potential for targeted marketing or pricing adjustments in underperforming regions.

- Between April and July, sales remained relatively stable within the £60,000–£65,000 range. This stability suggests a consistent customer base and predictable purchasing pattern, useful for budgeting and supply chain optimization.

2. Sales Performance Analysis (Product Value Mapping)



The Sales Performance Analysis dashboard visualizes the relationship between Quantity Sold and Total Sales using a bubble chart. This allows identification of high-performing and low-performing products across countries.

- Top Products: Laptops in the USA and T-shirts in the UK were identified as top-selling items with both high quantity and high revenue, positioning them as core revenue drivers.

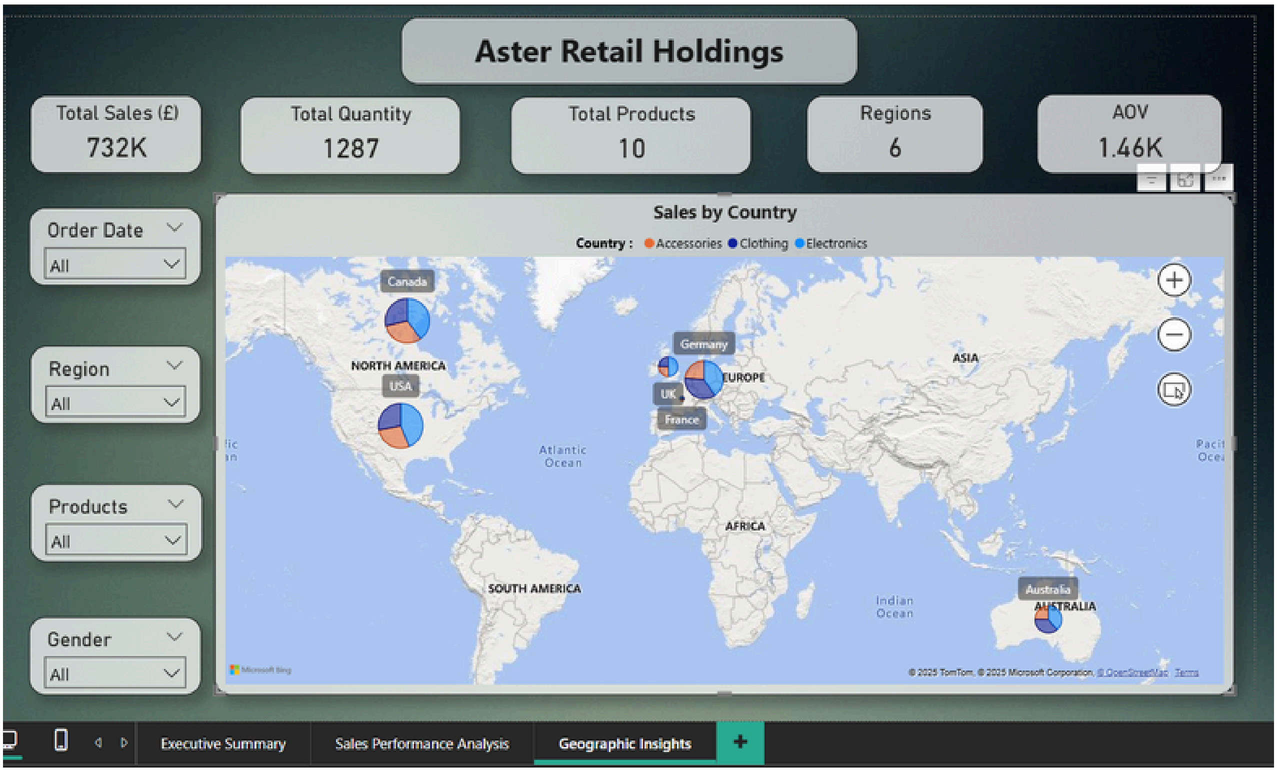
- Premium Products: Jackets and Watches in Canada emerged as premium products with high sales value but relatively lower quantities, indicating a luxury or niche market.

- Low Revenue, High Demand: The Tablets category in Germany showed strong demand but low revenue contribution, highlighting a potential area for pricing optimization or bundle strategies.

- Low Impact Products: Sunglasses in Australia fell into the low-demand, low-revenue quadrant, suggesting limited contribution to business performance.

- Outliers - Tablets in Canada appear as high-revenue outlier, suggesting they are premium-priced products with strong profitability. This indicates brand strength and margin efficiency, ideal for premium positioning or upselling strategies.

3. Geographic Insights (Regional Distribution)



The Geographic Insights page uses a Power BI Map Visual to depict Sales by Country, segmented by category. The visual highlights how different regions contribute to overall performance through an interactive, color-coded geographic view.

- Regional Leaders: The USA and Canada again dominated, with Electronics driving the majority of their sales.
- Balanced Markets: Germany showed balanced sales across all categories, reflecting a healthy product mix.
- Regional Gaps: Australia and UK showed moderate performance, while France lagged, potentially due to lower electronics demand.
- The USA and Canada together contribute over one-third of total global revenue (about 37.8%), indicating a heavy regional dependency. While this reflects strong brand presence in North America, it also poses risk concentration. A dip in demand from these regions could heavily impact overall sales stability.

8. Conclusion

The data analytics review of Aster Retail Holdings (London) for the year 2022 provides comprehensive insights into the company's overall sales performance, customer behaviour and product distribution. The analysis revealed consistent sales growth across major categories such as Electronics and Clothing, with Accessories showing moderate but stable performance. Regional analysis highlighted USA and Canada as top-performing markets, while certain regions displayed potential for targeted marketing and expansion. However, France recorded comparatively lower performance, likely due to reduced sales in the Electronics category, indicating an opportunity for category-specific promotional strategies in that region.

Customer segmentation and order behaviour patterns indicate varied purchasing trends across regions and product categories. However, a few outliers in sales and order volume indicate occasional bulk transactions or seasonal promotions that should be further examined to ensure inventory alignment and demand forecasting accuracy.

The Power BI dashboard translated raw sales data into meaningful business insights, helping management identify revenue patterns, evaluate product category performance and track sales variations across time. The 2022 analysis reveals that Aster Retail Holdings (London) maintained a consistent sales trend throughout the year, with a clear surge in August followed by a decline in December. These findings highlight the importance of seasonal demand planning and provide a strong basis for improving inventory control and future sales strategies.